

Methods in Java

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In Java, a method is used to do a specific work and helps make the program simple and reusable.

What is a method in Java?

A method is a block of code, or a collection of statements, or a set of code grouped to perform a certain task or operation. We write a method once and use it many times. We do not need to write code again and again. It also provides easy modification and readability of code just by adding or removing a chunk of code. The method is executed only when we call or invoke it.

The most important method in Java is the `main(String[] args)` method.

Advantages of using Methods

The following are the advantages of using methods in Java:

- **Reusability:** Methods provide code reusability. Once a method is defined, it can be called multiple times from various parts of a program.
- **Modularity:** Methods help breakdown complex problem into smaller manageable pieces.
- **Maintainability:** With methods, updating and debugging code becomes easier, since changes in one part of the program do not necessarily impact other parts.
- **Abstraction:** Method provide a way to encapsulate implementation details, exposing only the necessary parts to the user.

Method Naming Rules

While defining a method, remember that a method name must be a verb and start with a lower case letter. If the method name has more than two words, the first name must be a verb followed by an adjective or noun. In the multi word method name, the first letter of each word must be an uppercase except the first word. For example:

Single word, method name: `sum()`, `area()`

Multiword method name: `areaOfCircle()`, `endsWith()`

It is also possible that a method has the same name as another method in the same class. It is known as method overloading.

In String class we have:

```
substring(int start); //1-parameter
```

```
Substring(int start, int stop); //2-parameters
```

In method overloading the function names are same, but they differ with number of parameters, when data type of parameter is same.

```
package OOP;
```

```
/**
 * Write a description of class Shape here.
 *
 * @author (your name)
```

```

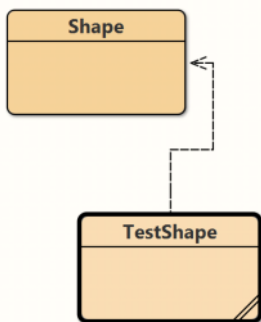
* @version (a version number or a date)
*/
public class Shape
{
    /**
     * @param len Radius of a Circle
     */
    public double area(double len){
        return len * Math.PI;
    }

    /**
     * @param len Length of the rectangle
     * @param bre Breadth of the rectangle
     */
    public double area(double len, double bre){
        return len * bre;
    }
}

package OOP;

/**
 * Write a description of class TestShap here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class TestShape
{
    public static void main(String[] args){
        Shape myshape = new Shape();
        System.out.print("\fArea of Rectangle: " +myshape.area(10, 5));
        System.out.print("\nArea of Square: " +myshape.area(10, 10));
        System.out.print("\nArea of Circle: " +myshape.area(10));
    }
}

```



Method Declaration:

The method declaration provides information about method attributes such as

visibility, return type, name, and arguments. It has six components that are known as method header Or prototype of the method.

```
public int sum(int a, int b)
{
    //method body
}
```

Access specifier
return type
method name
parameter list or formal parameter : receiver of the value.
method header or prototype.

```
sum(10, 5);
```

method signature
arguments or informal parameter or actual parameter (sender of the values).

Access Specifier

Access specifier or modifier is the access type of the method. It specifies the visibility of the method. Java provides four types of access specifiers:

- **public:** The method is accessible to all classes when we use the public specifier in our application.
- **private:** When we use a private access specifier, the method is accessible only in the classes in which it is defined.
- **protected:** When we use protected access specifier, the methods is accessible within the same package or subclasses in a different package.
- **default:** When we do not use any access specifier in the method declaration, Java uses the default access specifier by default. It is visible only from the same package.

Types of method:

- **Predefined method:** The methods that are already defined in the Java class libraries are known as predefined method. It is also known as the standard library method or built in method. We can directly use these methods just by calling them in the program at any point. For example: Math.Sqrt(), String class methods, length(), substring() etc.
- **User-defined method:** The method written by the user or programmer is known as User Defined Methods. These methods can be modified or customized according to the requirements.